

Project proposal

We will be doing a comparative analysis on different music generation models. We'll focus on generating single-track piano compositions. We will try to implement different architectures and compare their performances with a subjective evaluation based on a "Musical Turing Test".

1. Music Generation Taxonomy

1.1. Symbolic-domain vs Audio-domain

The first criteria to classify music generation models is what music representation to use.

Audio-domain means representing music as raw audio, encoded using the discrete fourier transform. This representation allows us to represent all possible sounds (if choosing a sampling frequency higher than human threshold), but also require huge amounts of data and computation.

Symbolic-domain means representing music using notes, events, or text tokens to represent melodies. These representations introduce a high inductive bias, because the model can focus directly on how melodies are created, and skip learning how frequencies are mixed to generate clean notes and chords. The advantage of this representation is that it is much more computationally efficient, and much smaller models can capture information of the data. This is the reason why we are going to use this type of representation. But it also comes with a downside, since we are constraining our model to generating simpler sounds, it will not be able to reproduce effects like sliding a string on the guitar, playing harmonics, or doing a vibrato on a violin.

1.2. Conditional vs unconditional

Another aspect to consider is how much control do we want to have over the music we generate. Conditional music generation allows us to generate music conditioned on a chord progression, a specific music genre, a sentiment, a prompt...

Unconditional music generation, on the other hand doesn't give us any control, and we can only generate random music, that will resemble our dataset.

In autoregressive models like RNN, we can slightly condition music giving it a starting melody, and ask it to continue it, but we cannot do this with all the models, for example an Autoencoder or a GAN.

As conditional generation requires more complex models and more complex datasets, since we now need labeled data, we will only implement unconditional models.

1.3. Multi-instrument vs single instrument

Some models can generate multi-track music, each track representing different instruments. But learning to synchronize all of the different instruments, both the chords the rhythm and the tempo is a really hard task, and thus we will stick with the single instrument piano tracks.

2. Timeline

12th April goals:

- Having a common interface to load the dataset and to convert midi outputs to audio.
- Have the 4 models implemented or at least started.

19th April goals:

- All models are implemented and optimized.
- Started creating a test poll to subjectively evaluate our models with a “Musical Turing Test”.
- First draft of the paper finished.

3. Models

As we have previously mentioned, we intend to implement 4 different models to generate quality piano compositions.

1. LSTM with Attention or Transformer
2. LSTM-VAE o CNN-VAE
3. CNN-GAN
4. LSTM-GAN

Since we want to try different architectures, evaluating the models against state-of-the-art is hard. In order to have a valid metric we need to use the same dataset, with the same music representation and the same loss function as the one we want to compare to, and this is not possible, since the architectures we chose have different loss functions.

Thus, we have decided to measure the quality of our models with a subjective poll, that will ask participants to mark different audio clips as human or ai-generated. With this we can determine if our music is coherent and “musical”.

4. Dataset

We will use the GiantMIDI-Piano dataset, since it's a well-known midi dataset containing solo piano containing 7,236 works composed by 1,787 composers.

5. Papers and articles

Mido: <https://readthedocs.org/projects/mido/downloads/pdf/latest/>

A Survey of AI Music Generation Tools and Models: <https://arxiv.org/pdf/2308.12982.pdf>

MidiNet: <https://arxiv.org/pdf/1703.10847.pdf>

DeepJ: <https://arxiv.org/pdf/1801.00887.pdf>

MusicVAE: <https://arxiv.org/pdf/1803.05428.pdf>

MuseGAN: <https://arxiv.org/pdf/1709.06298.pdf>

Music Transformer: <https://arxiv.org/pdf/1809.04281.pdf>

GAN evaluation: <https://machinelearningmastery.com/how-to-evaluate-generative-adversarial-networks/>

GiantMIDI-Piano: <https://arxiv.org/pdf/2010.07061.pdf>